CSE STUDY HUB

Emerging Technology

UNIT 2 · Cloud Computing & AI/Data Science

What this unit is about

This unit builds understanding of **Cloud Computing & AI/Data Science**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Cloud computing definition, characteristics and on-demand resources

2. Service models: IaaS, PaaS and SaaS

3. Deployment models: public, private, hybrid and community cloud

4. Virtualisation, containers, scalability and elasticity

5. AI, machine learning and data science overview

6. Data lifecycle: collection, cleaning, analysis, modelling and visualisation

7. Cloud applications, benefits, costs, security and vendor considerations

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 2 NOTES · PART 1

Cloud computing definition, characteristics and on-demand resources

Cloud computing provides on-demand shared computing resources over a network. IaaS supplies infrastructure, PaaS supplies a development platform, and SaaS supplies a ready-to-use application.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Service models: IaaS, PaaS and SaaS

Treat Service models: IaaS, PaaS and SaaS as a structured model. Name its components in order, explain the responsibility of each component, and show how information moves between them. Include a labelled block diagram and one use case.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Deployment models: public, private, hybrid and community cloud

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Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Cloud Computing & AI/Data Science, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 2

Virtualisation, containers, scalability and elasticity

Virtualisation, containers, scalability and elasticity is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

AI, machine learning and data science overview

Start with a precise definition of AI, machine learning and data science overview. Explain its purpose, the problem it solves, its main components and where it is used. In an exam answer, add one practical example and distinguish it from the nearest related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Cloud Computing & AI/Data Science, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 3

Data lifecycle: collection, cleaning, analysis, modelling and visualisation

Present Data lifecycle: collection, cleaning, analysis, modelling and visualisation as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Cloud applications, benefits, costs, security and vendor considerations

Cloud computing provides on-demand shared computing resources over a network. IaaS supplies infrastructure, PaaS supplies a development platform, and SaaS supplies a ready-to-use application.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Cloud Computing & AI/Data Science, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Cloud computing definition, characteristics and on-demand resources** — know the definition, core working and one example.

• **Service models: IaaS, PaaS and SaaS** — know the definition, core working and one example.

• **Deployment models: public, private, hybrid and community cloud** — know the definition, core working and one example.

• **Virtualisation, containers, scalability and elasticity** — know the definition, core working and one example.

• **AI, machine learning and data science overview** — know the definition, core working and one example.

• **Data lifecycle: collection, cleaning, analysis, modelling and visualisation** — know the definition, core working and one example.

• **Cloud applications, benefits, costs, security and vendor considerations** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Cloud Computing & AI/Data Science. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Cloud computing definition, characteristics and on-demand resources and explain its importance.

Q2. Explain the working/steps involved in Service models: IaaS, PaaS and SaaS.

Q3. Differentiate between two related concepts from Cloud Computing & AI/Data Science.

Q4. Give one practical application of Cloud applications, benefits, costs, security and vendor considerations and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Cloud Computing & AI/Data Science.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.